Autonomous Real Estate

Sales Funnel Agent

Amarras San Pedro · Buenos Aires - Argentina

THE PAST

THE PAST

THE PROBLEM

My parents sell waterfront lots. They answer the same questions every day.

Manual follow-up

Every lead needs a personal call or WhatsApp — no automation, no system

Repetitive questions

"Price?" "Lot size?" — answered 10× per day manually

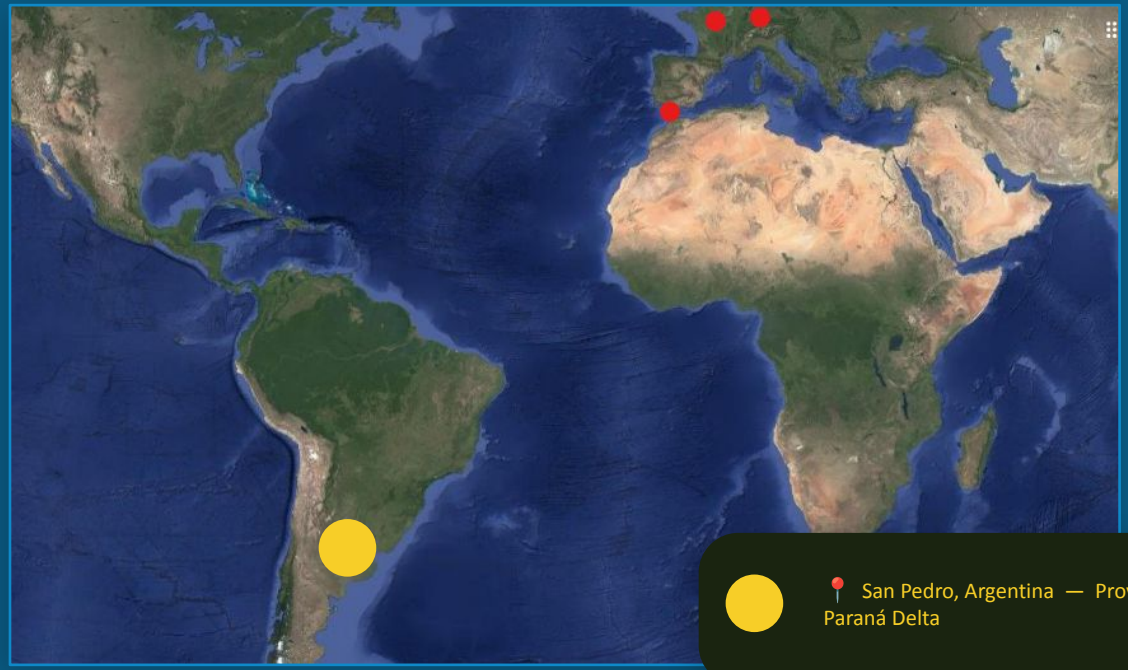
No outbound reach

No way to systematically find and contact agencies or investors

No marketing

No digital presence, no lead capture, no way to track prospects

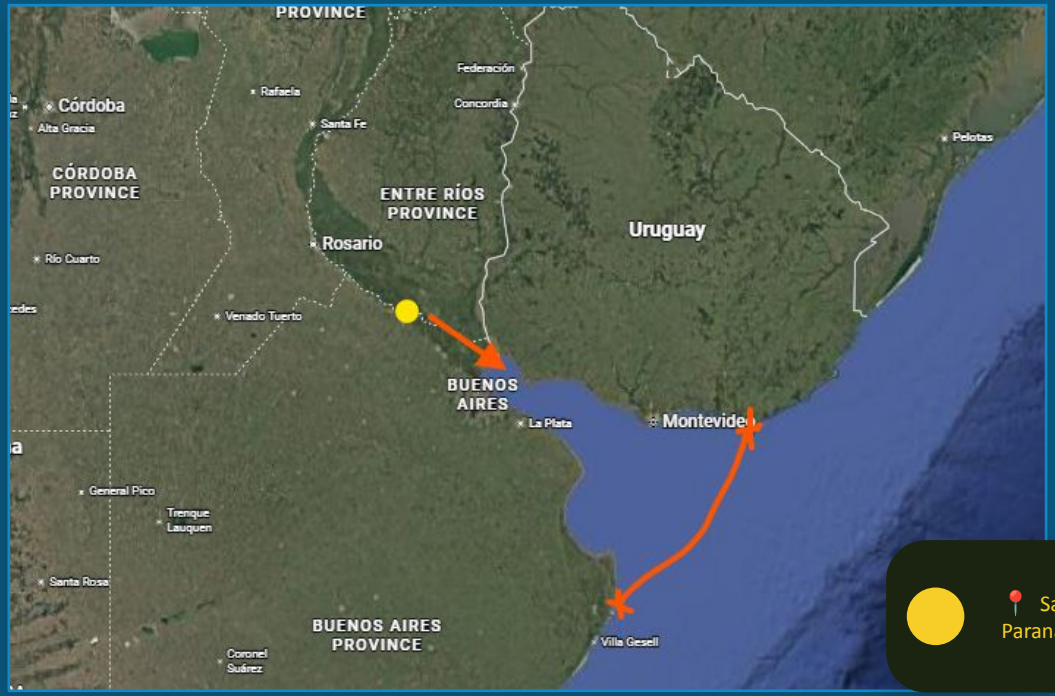
WHERE IN THE WORLD



📍 San Pedro, Argentina — Province of Buenos Aires · Paraná Delta

70 km north of Buenos Aires · Paraná River waterfront

THE PROJECT LOCATION



📍 San Pedro, Argentina — Province of Buenos Aires · Paraná Delta

70 km north of Buenos Aires · Paraná River waterfront

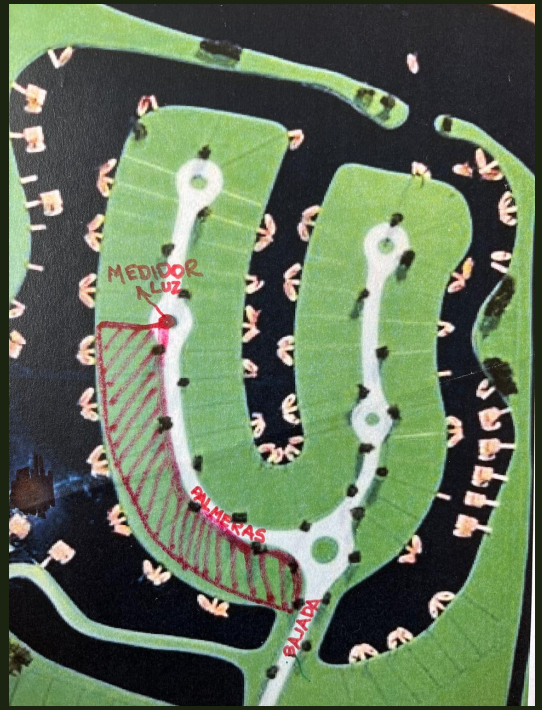
THE PROJECT LOCATION

Amarras San Pedro

- San Pedro, Bs. As.
- Paraná River waterfront
- Lots: 544 – 1,188 m²
- Price: U$S 120/m²
- 70 km from CABA
- Club Náutico Boating I
- Gated community

Waterfront living · Nature · Exclusivity · 70 km from Buenos Aires

AMARRAS SAN PEDRO



Replace placeholders with real property photos

AMARRAS SAN PEDRO

Replace placeholders with real property photos

THE SOLUTION

A dual-mode autonomous agent

🏠 INBOUND ENGINE

→ Answers buyer questions 24/7

→ Pulls from real property docs (RAG)

→ Classifies leads: HOT / WARM / COLD

→ Fires Telegram alert for HOT leads

→ Sends HTML email for WARM leads

→ Logs Warm and Hot Leads every session to Google Sheets

+

🔍 OUTBOUND ENGINE

→ Searches agencies & investors (Tavily)

→ Extracts emails & phones (Hunter.io)

→ Auto-logs contacts to Google Sheets to a classification prospects sheets

→ Export Prospect Intelligence Report

→ Keyword-powered quick searches

→ Click-to-email directly from the UI

One platform · Two revenue levers · Zero manual effort

Built with Module 3 tools & 7 API Integrations

Agent Framework

LangChain + LangGraph

ReAct Pattern

create_react_agent + 3 @tool functions

Vector Database

Pinecone · text-embedding-3-large · 69 vectors

APIs / Tools

Tavily · Hunter.io · Pinecone RAG

Notifications

Telegram Bot · Gmail SMTP · Google Sheets

Orchestration

n8n webhook · auto-log every conversation

Deploy

Railway (live) · Flask · Python 3.12

Version Control

Git / GitHub · 30+ meaningful commits

ARCHITECTURE

The refactor that halved response time

BEFORE

retrieve ctx › react agent › classify lead › telegram › gmail › output

~7 seconds ⚠️ user waits for everything

ASYNC REFACTOR ↓

AFTER

retrieve ctx › react agent › output

~3.5 s ✅ JSON to user immediately

🔄 **Background thread (user doesn't wait)**

classify lead › telegram (HOT) › gmail (WARM) › n8n → Sheets

Separate what the user waits for from what can run silently in the background

RAG + LEAD INTELLIGENCE

📚 RAG KNOWLEDGE BASE

Pinecone index:	'n8n' · 1024 dims
Embedding:	text-embedding-3-large
Vectors:	69 chunks · 2 clean docs
Similarity:	cosine ≥ 0.35
Data fix:	Removed conflicting PDFs
Ground truth:	Verified prices in system prompt

🎯 LEAD CLASSIFIER

🔥 HOT

Buying intent · wants contact · asks about payment

🌡️ WARM

General interest · price questions · active browsing

🧊 COLD

Vague · no intent · one-off casual question

gpt-4o-mini · temp=0 · async thread · JSON output

Data quality > model complexity · Cleaning source docs beat every algorithm change

THE QR CODE FOR THE WEB APP

LIVE DEMO

1 Language picker

Select EN or ES
Locked for the session

2 Ask about lots

"What lots are available?"
RAG retrieves docs instantly

3 Trigger HOT lead

"I want to buy one"
CTA button appears in response

4 Outbound research

Search agencies → find emails
→ export report

5 Google Sheets live

Show auto-populated
lead log in real time

6 Telegram alert

HOT notification fired
silently in background

Live on Railway · Tested EN + ES · 2 happy first customers (my parents)

CHALLENGES & LEARNINGS

The N8N pivot

Planned N8N as backbone. Realized Day 1–2 it couldn't handle Python-heavy logic. Rebuilt mental model mid-project.

💡 *N8N is best as a thin notification layer, not an orchestrator.*

Response latency

6 sequential nodes = 7s. Two LLM calls + SMTP + Telegram all blocking the user.

💡 *Async refactor: 3-node fast path + background thread → 3.5s.*

RAG data quality

Conflicting pricing in source PDFs. Agent gave wrong prices. Algorithm changes didn't help.

💡 *Data cleaning + hardcoded ground truth in system prompt fixed it.*

I wouldn't change the mistakes. They were necessary.

WHAT'S NEXT

Buy a domain

Embed agent into the Amarras San Pedro website

Mobile app

Wrap Railway URL into a native iOS/Android app

Lead capture form

CTA button → collect name + email + phone

Voice bot

Real-time voice conversation agent (v2)

Lot map

Visual layout — which lots are still available

CRM dashboard

Lead history over time · conversion tracking

"It's deployed. My parents are using it. There's still a lot to do — and I know exactly what."

THANK YOU!

ANY QUESTIONS?